Dataset & Remote Sensing Basics

- Can you explain what Sentinel-2 imagery is and why it has multiple spectral bands?

  - It's a satellite mission by European Space Agency comprised of two satellites that capture high resolution optical imagery from space.
  - It has multiple spectral bands because each band belongs to a different wavelength giving it the ability to view images in a variety of ways.
- What's the difference between an RGB image and a multi-band remote sensing image?
  - An RGB image only has 3 color channels (Red, Green, and Blue) while a multi-band remote sensing image has 12 total spectral bands ranging from visible light to shortwave infrared.
- Why might we use indices like NDVI instead of just RGB values?

  - Indices like NDVI are normalized numerical equations that we use so that computers can interpret the vegetation health from a satellite image. We use them instead of RGB values since computers can't calculate or deduct how healthy the vegetation is from a single color represented by a number.

Data Exploration

- When you visualized the images, what patterns or differences did you notice across bands?

  - The spots in the image that showed more vegetation were brighter than spots with less or close to no vegetation.
- How are labels structured in this dataset (classification, segmentation mask, bounding box, etc.)?
  - These labels are structured as a segmentation map where each pixel in the image has its own NDVI value.
- If an image file and its label file don't match up, what kind of issues would that cause in ML?

  - The machine would output wrong information about the image causing the data to be invalid.

Data Characteristics

- What does “resolution” mean in remote sensing images, and why does it matter for analysis?

  - It refers to the size of each pixel. Higher resolution means finer detail which leads to more accurate data.
- How can metadata (like acquisition date or sensor type) affect the performance of a model?
  - It can both positively and negatively affect the model. On one hand, it will provide additional context training the model to recognize even more patterns. On the other, there will be more room for error in the case that it interprets that extra information as things to constantly check for in other samples.

- What problems might missing labels or noisy data introduce in training?

  - Missing labels in the dataset will cause mismatched data making the outputs from the model inconsistent and wrong. Noisy data will make the model work harder and reduces efficiency.

Preprocessing

- Why do we normalize images before feeding them into a machine learning model?

  - It puts the images to the same scale therefore making them more consistent and easier to visualize for us and the model.
- What’s the difference between min–max normalization and standardization?
  - Min-max normalization puts scales down the values you want to test between 0 and 1. Standardization helps visualize how close or far away a certain element is from the median.
- Why do we need to encode categorical labels as numbers?
  - It helps instantly know the size of our dataset and makes it easier to search for a specific one.
- How does resizing images affect both computational efficiency and model accuracy?

  - If you go too large you lose computational efficiency because it takes up more memory space, but model accuracy increases since you get finer detail.

Machine Learning Fundamentals

- What's the difference between features and labels in supervised learning?

  - Features are the independent variables or inputs that we give the model to base the training on. Labels are the dependent variables or the outputs that we want the model to try to predict.
- Why do we split data into training and test (and sometimes validation) sets?
  - The training set is used to feed the model new information and let it recognize patterns so that when it sees the test set for the very first time it should know what to do. Validation sets are used in between these two so that we can make sure the model is recognizing patterns instead of memorizing the training data.
- Can you explain the difference between classification and regression tasks?
  - Classification task is when we want a model to predict what label belongs to an element. Regression task is when the model predicts numerical values.
- If your dataset is imbalanced (e.g., lots of "forest" but few "water" samples), what problems might that cause?
  - The model will always expect there to be more forest than water and would give inaccurate data when tested with a brand-new dataset.
- How would you evaluate whether your model is learning useful patterns vs just memorizing the training data?
  - I would give it a dataset it has yet to see and observe how it behaves.
- What role does overfitting play when working with high-resolution images?

  - It is bound to have more overfitting in high-resolution images since there are more pixels the model could memorize.

Additional Findings

Libraries:

- Rasterio
  - Used to read TIFF satellite images
- Matplotlib
  - Used to visualize the images
- Numpy
  - Handles numerical math.

Metadata

- Resolution: 256 × 256
- File name: S2A_MSIL2A_20171025T150721_N0500_R082_T19QGA_20230910T203157.SAFE_ndvi_0.tiff
  1 2 3 4 5 6 7 8 9 10

1. Satellite name: Sentinel-2A
2. Processing level: MSI (MultiSpectral Instrument) = Sensor onboard
   L2A (Level-2A product) = Data is ready for vegetation analysis
3. Date captured: YYYYMMDDTHHMMSS ← Format (UTC)
   year/month/day | hour:min:sec
4. Processing baseline number: N0500 = Software version used
5. Relative orbit number: R082 = Orbit 82
6. Tile ID: T = Tile
   (MGRS) 19 = UTM zone 19
   QGA = specific 100km grid square
7. Product processed date: Same format as day captured
8. ESA product container format: .SAFE = standard folder structure for Sentinel products
9. Classification: ndvi = Label
   img = Sample
10. Index number
11. File format